

G Ensemble

A divide-down string machine with a three-phase ensemble.



Overview

G Ensemble is a string machine in the spirit of the classic 70s string synths. Like the originals, it uses twelve master oscillators and divides them down for every octave, so every key can sound at once, nothing is ever cut off, and all the notes of the same name are locked in phase.

Four registers are mixed together: VIOLIN (a bright 8'), VIOLA (a mellow 8'), CELLO (16') and BASS (16' plus 32', on the lower keys only). Each key has its own simple envelope: ATTACK fades it in, RELEASE lets it ring on after you let go.

The magic is the ENSEMBLE: three delay lines swept by a slow and a fast wobble, 120 degrees apart. It turns the plain, buzzy oscillators into the lush, shimmering string sound.

Reach for it for

- Lush 70s and 80s string pads
- Slow swells and evolving backgrounds
- Disco string stabs
- Dark cello and bass drones
- Wide, shimmering layers under other keyboards

Playing it

G Ensemble is an instrument: it has no audio input and makes its sound from MIDI.

1. Insert G Ensemble on a track.
2. Arm the track for recording and set its input to your MIDI keyboard (or draw notes in a MIDI item).
3. Play. The MIDI IN light glows while notes are sounding, and the display shows how many keys are sounding.
4. If you use a multi-channel MIDI setup, set CHANNEL in the MIDI section; OMNI listens to all channels.

There is no limit on how many keys sound at once. It responds to all-notes-off (CC120 and CC123); incoming MIDI is passed on down the chain.

Controls

The header

PRESETS, arrows and display	PRESETS (or a click on the display) opens the list of factory presets; the ◀ / ▶ keys step through them. The display shows the preset number and name. A preset sets the STRINGS, ENSEMBLE and TONE controls; TUNE, the MIDI channel, DRIFT, OUTPUT, SAT, the output stage and the stereo controls are left as they are. The last entry resets everything to the defaults.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.

?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.
PANIC	Stops every key at once.

The display

Shows the three ensemble wobbles moving, ENSEMBLE THREE-PHASE (or ENSEMBLE OFF), and KEYS: the number of keys sounding.

STRINGS

VIOLIN	The level of the bright 8' register, 0 - 100%.
VIOLA	The level of the mellow 8' register, 0 - 100%: softer and more hollow than the violin.
CELLO	The level of the 16' register (an octave down), 0 - 100%.
BASS	The level of the 16' and 32' bass register, 0 - 100%. It plays only on the keys below C3 (MIDI note 48).
ATTACK	How long each key takes to fade in, 5 - 2000 ms. Short for stabs, long for swells.
RELEASE	How long each key rings on after you let go, 50 - 5000 ms.
VELOCITY	How much your playing dynamics matter, 0 - 100%. At 0 (default) every note plays at the same level, like an organ, as on the originals.

ENSEMBLE

ON	Switches the ensemble on or off. Off gives the dry, buzzy oscillators.
DEPTH	How far the three delay lines are swept, 0 - 100%. More depth gives more shimmer and detune.

SPEED	The speed of both wobbles, 25 - 200% of the standard rate (100%).
WIDTH	The stereo width of the ensemble, 0 - 100%. At 0 it is mono.

VOICE

TONE	A low-pass tone control on the whole sound, 0 - 100%. Lower is darker.
TUNE	Fine tuning, -50 to +50 cents.

DRIFT

DRIFT	A slow random wander, 0 - 100% (25% by default). It moves the pitch of the whole instrument (up to +/-20 cents at 100%), the TONE, and the speed of the ensemble. The keys stay locked together, as on a divide-down machine.
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MIDI

CHANNEL	The MIDI channel G Ensemble listens to: OMNI (all) or 1 - 16.
MIDI IN	Lights when a note arrives and while keys are sounding. The number underneath counts the keys sounding.

MASTER

SAT	Band saturation on the way out, 0 - 100%: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.
OFF / CLIP / LIMIT	The final stage: nothing, a soft clipper that holds the peaks at -0.3 dBFS (no latency), or a lookahead limiter that holds them at -0.3 dBFS (3 ms latency, compensated by REAPER). The CLIP key flashes red while the clipper is working.

MONO	Makes everything below the set frequency mono, 20 - 500 Hz (fully left = OFF). Use it to keep the low end centred.
WIDTH	Widens the stereo image above the MONO frequency, from +0% up to +100% (twice the side signal).
OUTPUT	The output level, -24 to +12 dB.
OUT meters	Output level, left and right. Click a meter to clear its peak hold.

Presets

Classic Strings	The default: all four registers, a 150 ms attack, a 0.9 s release and the ensemble on.
Violins Only	Just the violin register, a quick attack, a deep, slightly faster and fully wide ensemble, brighter tone.
Dark Cellos	Mostly viola, cello and bass with little violin, a slow attack, a long release and a darker tone.
Space Pad	All registers with a slow 0.9 s attack, a 4 s release and a deep, slow, fully wide ensemble.
Disco Stabs	Violin and viola with a fast attack and short release, and some velocity response for accents.
Full Orchestra-ish	All four registers with plenty of bass and cello, a medium attack and a long release.
Dry (no ensemble)	The default registers with the ensemble off.
Slow Swell	Violin, viola and a little cello with a 1.2 s attack, a 5 s release and the deepest, slowest ensemble.
Reset all to defaults	Every control back to its starting value, including TUNE, the MIDI channel, DRIFT, OUTPUT, SAT, the output stage and the stereo controls.

Tips

- Hold a chord and switch the ensemble ON and off to hear what it does - most of the sound is in the ensemble.
- Keep BASS for the lowest part of your part; it only plays below C3, so it adds weight without muddying the chords.
- For a pad that blooms, set ATTACK to 500 ms or more and RELEASE to 2 s or more, and overlap the chords.
- If the strings fight with other parts, lower TONE rather than turning them down.

File: gfx-g-ensemble.jsfx